$a_n = c a_{n-1} + d r^n \quad , a_0 = b$

$a_n^{(p)} = A r^n \rightarrow$ should also satisfy base case $\Rightarrow$

$A r^n \rightarrow$ after opening the recurrence, $A r^n$ is the one dominating,

so, $a_n^{(p)}$ assume $= A r^n$.

to get unique solution $\Rightarrow$ & satisfy base case

If $r \neq c$ $\quad a_n = a_n^{(h)} + a_n^{(p)}$.

$c_{n+2}a_{n+2} + c_{n+1}a_{n+1} + c_n a_n + d \cdot n = 0$ $\quad a_0 = b_0,\ a_1 = b_1$

opening them up → decreasing powers, $n+2, n+1, n$

r^n → dominating

$a_n^{(p)} = Ar^n$

$a_n^{(h)} =$ so of $c_{n+2}a_{n+2} + c_{n+1}a_{n+1} + c_n a_n = 0$.

don't want, $a_n^{(p)}$ to common to $a_n^{(h)}$.

$a_n = \lambda x^n$

$c_{n+2}x^2 + c_{n+1}x + c_n = 0$

Real, distinct	complex	Real, equal
r_1, r_2	θ_1, θ_2	r_1
$a_n = \lambda_1 r_1^n + \lambda_2 r_2^n$	$\lambda_1 \theta_1^n + \lambda_2 \theta_2^n$	problem in boundary condition

$\lambda_1 r_1^n + \lambda_2 n r_1^n$ → smallest polynomial that fits eqn.

Real distinct:

1. $r_1 \neq r_2 \neq r$

a. $a_n = Ar^n + \lambda_1 r_1^n + \lambda_2 r_2^n$

2. $r = r_1$ or $r = r_2$.

An → consumed by homogeneous.

$= An r_1^n + \lambda_1 r_1^n + \lambda_2 r_2^n$

complex: → Always (1). $\theta_1 \neq \theta_2 \neq r$ because r is real ~~part~~...

$\theta_1 = r^n(\cos(n\theta) + i\sin n\theta)$

will take n is $a_n^{(p)}$ if creates conflict.

Real equal:

a. $r_1 \neq r$

$a_n = a_n^{(p)} + a_n^{(h)}$

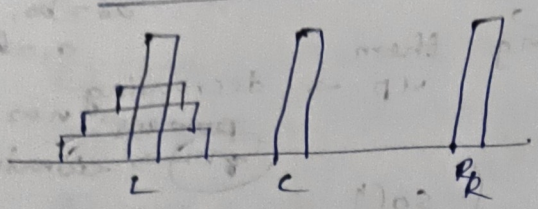
$= \lambda r^n + \lambda_1 r_1^n + \lambda_2 n \cdot r_1^n$.

b. $a_n = An^2 r^n + \lambda_1 r_1^n + \lambda_2 n r_1^n$

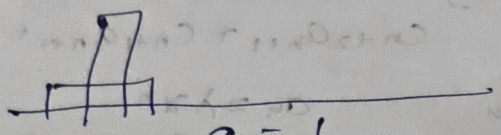
→ Constant from homogeneous part, combined with particular part, satisfies boundary ~~given~~ taken 2 and respond $a_n^{(p)}$.

So,

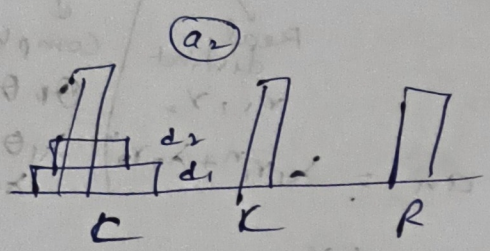
Tower of Hanoi:



a_n = # moves required for n disks to move from L → R

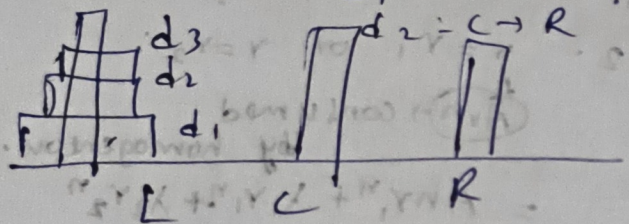


$a_1 = 1$



$d_2 : L \to C$

$d_1 : L \to R$



$\{d_2 \cdots d_n\}$: $\{d_2, d_3\} : L \to C$

d_1 : $d_1 : L \to R$

$\{d_2 \cdots d_n\}$ $\{d_2, d_3\} C \to R$

$a_n = 2a_{n-1} + 1$

$a_n^{(p)} = A(1)^n$

$\lambda = -1$

$a_n^{(h)} = \lambda \cdot 2^n$

$a_n = \lambda \cdot 2^n - 1$

$n = 1$

$2\lambda - 1 = 1$

$\lambda = 1$

$\boxed{a_n = 2^n - 1}$

~~n length str numb~~

n digit ~~number~~ string, each digit {0, 2}

a_n = # n digit string where even no. of 1s appear.

$\boxed{a_1 = 9}$

$$a_n \begin{cases} (10^{n-1} - a_{n-1}), & \text{last digit } 1 \\ (9a_{n-1}), & \text{last digit is not 1.} \end{cases}$$

$a_n = 8a_{n-1} + (10^{n-1})$

$a_n^{(h)} = \lambda(8)^n$

$a_n^{(p)} = A(10^n - 1)$

$10^n \; 8^n$ no correlation

$A(10^n - 1) = 8(A10^{n-2}) + 10^{n-1}$

$A = \frac{8A}{10} + 1$

$\Rightarrow A = 5$

$a_n^{(p)} = 5\left(\frac{1}{2}10^n\right)$

$a_n^{(h)} = \lambda(8)^n$

$5 + \lambda(8) = 9$

$\lambda = \frac{1}{2}$

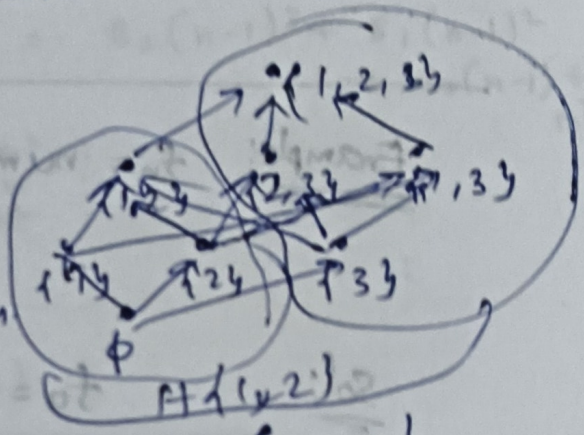
$a_n = \frac{1}{2}(10^n + 8^n)$

House diagram.

$\{1, 2, \ldots, n\} \subseteq$

a_n = # edges in the House diagram

$a_1 = 1$

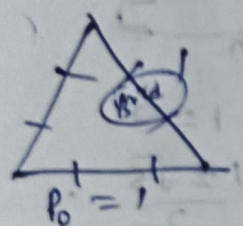


$a_n = 2a_{n-1} + (2^{n-1})$ derived

$a_n^{(p)} = An2^{n-1}$

$a_n^{(h)} = \lambda \cdot 2^n$ → from have ex.

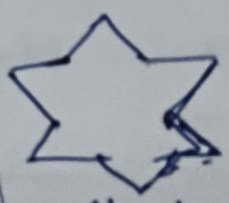
$a_n = a_n^{(h)} + a_n^{(p)} = \lambda 2^n + An2^{n-1}$



Side length = 1

sides = 3

area = $\frac{\sqrt{3}}{4}$

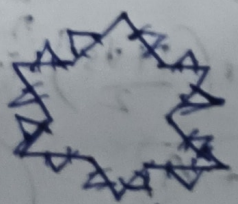


P_1

Side length = $\frac{1}{3}$

sides = 4.3

area = $\frac{\sqrt{3}}{4} + 3\left(\frac{\sqrt{3}}{4}\right)\left(\frac{1}{3}\right)^2$



Side length $\left(\frac{1}{3}\right)^2$

sides = $(4)^2 \cdot 3$

area = area(P_1) + $+ (4.3 \cdot) \frac{\sqrt{3}}{4}\left(\frac{1}{3}\cdot\right)^2$

P_n

Side length $= \left(\frac{1}{3}\right)^n$

sides $= (4)^n \cdot 3$

area $=$ (area)$_{P_{n-1}}$ + $(4^{n-1})3 \cdot \frac{\sqrt{3}}{4}\left(\frac{1}{3^{2n}}\right)$

$a_n = a_{n-1} + (4^{n-1} \cdot 3) \frac{\sqrt{3}}{4} \frac{1}{3^{2n}}$

$a_n = a_{n-1} + \frac{3\sqrt{3}}{4} \frac{4^{n-1}}{3^{2n}}$

$a_n = a_{n-1} + \frac{3\sqrt{3}}{16}\left(\frac{4}{9}\right)^n$

$a_n^{(h)} = \lambda$

$a_n^{(p)} = A\left(\frac{4}{9}\right)^n$

$\Rightarrow A\left[\left(\frac{4}{9}\right)^{n-1}\right][$

Example: f_n = number of additions required to obtain nth Fib. no.

$F_n = F_{n-1} + F_{n-2}$, $F_0 = 0, F_1 = 1$

Sol: $f_0 = 0$, $f_1 = 0$

(no meaning)

$f_n = f_{n-1} + f_{n-2} + 1$

$a_n^{(p)} = A \cdot (1)^n = A$

$A = A + A + 1$

$A = -1$

$a_n^{(h)}$

$f_n = f_{n-1} + f_{n-2}$

$f_n = \lambda r^n$

$r^2 - r - 1 = 0$

$r = \frac{1 \pm \sqrt{5}}{2}$

$f_n = \lambda_1 r_1^n + \lambda_2 r_2^n$

$= \lambda_1\left(\frac{1+\sqrt{5}}{2}\right)^n + \lambda_2\left(\frac{1-\sqrt{5}}{2}\right)^n$

$f_n = \lambda_1 r_1^n + \lambda_2 r_2^n + A$

$f_n = \lambda_1 r_1^n + \lambda_2 r_2^n - 1$

direct way of solving

next, Ex, generating